

Mount Maunganui Reef – Amalgamating Design and the Constraints of Construction

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Abstract

Mount Maunganui Reef, on the north eastern coast of New Zealand, has been built from sand-filled geotextile containers. The reef was specifically designed to enhance the surfing amenity at Mount Maunganui Beach. Computer modelling was utilized to develop a reef design for the local wave climate. The final computer design resulted in a 'Delta' reef that would shoal incoming waves into a sharp breaking peak, or 'A-frame', and then peel fast and hollow for over 50 m in each direction, producing intense breaking surfing waves for competent surfers. While the iterative computer design approach produced a reef shape for the site that optimized the typical surfing wave climate (i.e. 1-1.5 m straight-crested waves with 8-10 sec period), the reality of replicating this exact shape on the seabed was constrained by the limits of construction materials and methodology. In the first instance, scaled physical modelling was used to amalgamate the final computer design with the construction units, i.e. geotextile containers of up to 60 m long by 7.2 m wide and 3.2 m high (container volumes of up to 900 m³). The second stage was to ensure that the geotextile containers could be accurately positioned on the seabed in the configuration developed during the physical modelling. A novel method was developed to deploy multiple geotextile containers with sub-meter accuracy and was implemented successfully. As a result, the Mount Maunganui reef has consistently produced high-quality surfing waves, even when prior to completion. The characteristics of the waves breaking on the Mount Maunganui reef have proven the reef design and verified the results of both the physical and numerical modelling. This paper describes some of the novel design and construction processes that were applied to construct the Mount Maunganui surfing reef, and the results of the surfing waves that the reef as produced.

1 Background

The Mount Maunganui Reef was first considered in 1996. At this stage it was considered an 'exercise' to determine what would be required for Resource Consent application for building permits as part of a PhD thesis (Mead, 2000). A desktop investigation of potential sites within 3 hrs of the University of Waikato was first undertaken. Site selection was achieved by comparing several alternative sites using a point scoring system that considered the following factors:

- access and distance from the University
- sources of construction materials
- logistics of construction and access
- existing modifications to adjacent shoreline
- perceived erosion problems
- existing surfing usage

Each factor was scored 1-5 with 5 being the positive – the highest score indicating the most preferable site. From this selection process, Mount Maunganui was clearly rated the highest.

Mount Maunganui is often considered New Zealand's 'Surf-City' and local interest in a surfing reef was very high. Together with interest from other University Departments (e.g. biology and social sciences), there was a drive to develop the academic exercise into a reality. The existing surf culture at Mount Maunganui and links between the University of Waikato, the local port (Port of Tauranga) and local tertiary education institution (Bay of Plenty Polytech) were all additional drivers that resulted in the development of the project. Following a range of investigations, public consultation and design works, the Resource Consents

application was filed in 1998 (Black et al., 1998; Mead et al., 1998).

Resource Consents for construction of the reef were granted through a non-notified process (i.e. without the need for a public hearing). However, as a result, the time-share facility in close proximity to the proposed reef site, Sun Pacific Villas, decided to take court action to stop the project proceeding – the Body Corporate was concerned that more unruly surfer-types would spoil their coastal location and increase the trespassing in the spa area during the winter months.

The consenting authority, Environment Bay of Plenty, and the University of Waikato won the case, and were awarded costs. However, the project lost momentum due to the long court delays.

In 2003, the Mount Reef Trust was formed in 2003 and employed an executive officer and professional funds-raiser. Sufficient funds were raised to commence construction in November 2005.

This paper describes the how the detailed computer design of the Mount Maunganui Reef was transformed into a structural design through physical modelling and novel construction techniques. In addition, the resulting surfing waves that the Mount Reef has produced are used to verify the successful design and construction process.

2 Computer Design

Initial design of the Mount Maunganui Reef was undertaken for the Resource Consent application (Mead et al., 1998). A comprehensive numerical modelling approach was undertaken which included

utilization of 4 years of offshore wave data, with transformation to the reef site at Tay Street, Mount Maunganui (Figure 1), in order to develop an inshore wave climate.

The design wave parameters for the reef were: wave height of 1-3 m; periods of 5-13 s with optimal period of 11 s, optimal wave approach angle of $50^\circ \pm 20^\circ$ True.



Figure 1. Map showing the location of Tay Street, Mount Maunganui, on New Zealand's north eastern coast

WBEND (Wave BENDING) and Boussinesq model 3DD, from the 3DD Suite of numerical models (Black, 2001) were utilised for reef design and functional/impact assessment, both for resource consent application (Mead et al., 1998) and detailed design (Black and Mead, 2005).

The model WBEND (Black and Rosenberg, 1992) has been applied to a broad range of physical environments and has been calibrated and shown to be accurate on many occasions. The model has also been successfully validated and used as one of the primary design tools for the multi-purpose reef designed for the Gold Coast (Black et al., 1998), validated against surfing reef measurements (e.g. Mead and Black, 2001a, b), laboratory data (Black and Rosenberg, 1992) and in open coast conditions (Black *et al.*, 1998). In addition, significant modifications have recently been applied to model WBEND to simulate diffraction over complex bathymetric features (Black, unpub).

In 2004, final reef design work was undertaken, and due to the directionally variable wave climate at the Mount, a 'Delta' type reef was designed that would break waves from a single peak (an A-frame) and peel them in each direction for 50-65 m (Figure 2).

A database of over 40 mostly world-class surfing breaks was used to determine the peel angles and reef gradients in the Mount Reef design (Mead, 2000). The final reef design is symmetrical about the central axis and consists of 2 zones (Figure 2). There is a leading offshore "focus zone" which is low amplitude (with

gradient of 1:20 to 1:50). This is backed by a pointed horseshoe-shaped breaking zone (with gradient of 1:20 to 1:10) (Black and Mead, 2005). The latter gradients on the breaking component of the reef, the 'wedge' (Mead and Black, 2001a), was designed to break waves with a low peel angle (i.e. fast peeling) and a high breaking intensity (i.e. hollow or 'tubing') – requiring skill levels of competent to advanced to successfully ride (Hutt, 1997).

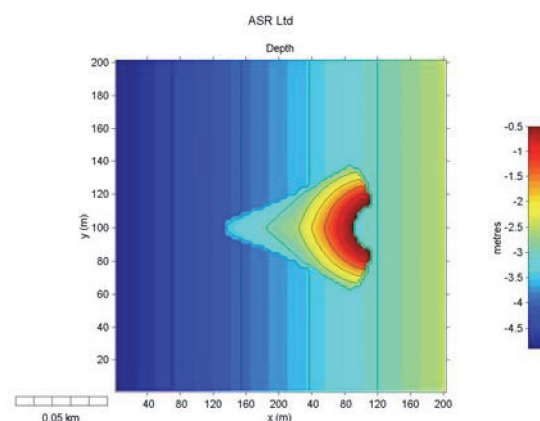


Figure 2. The final 'Delta' type reef design for Mount Maunganui – the bathymetry has been rotated for modelling purposes so that waves are approaching from the left of the picture and the beach is on the right.

In the design shown (Figure 2), the breaking distance is a minimum of 50 m on each side, but ride length increases with increasing swell height. The lower level acts as a focus but also acts to condition the waves by starting the shoaling process before reaching the breaking zone (Mead and Black, 2001b).

Once finalised, the detailed computer reef design was used as the basis for physical modelling in order to amalgamate the design with the restrictions of construction materials and methods.

3 Physical Modelling

Physical modelling was undertaken in ASR's 8.5 m x 4.5 m wave basin, with maximum water depths up to 0.6 m. Waves are generated from a water-filled compartment spanning the width of the pool. A heavy steel plunger is raised and lowered using an electric winch to form the waves. The generator is operated using a control box which can switch between manual and automatic and allows for the adjustment of timing, both the starting and holding of the generator at preset levels. This allows adjustments to the wave height and period. Wave heights up to 0.20 m can be easily generated.

The first simulations were undertaken using a 1:20 scale model of the computer-designed Mount Reef (Figure 3).

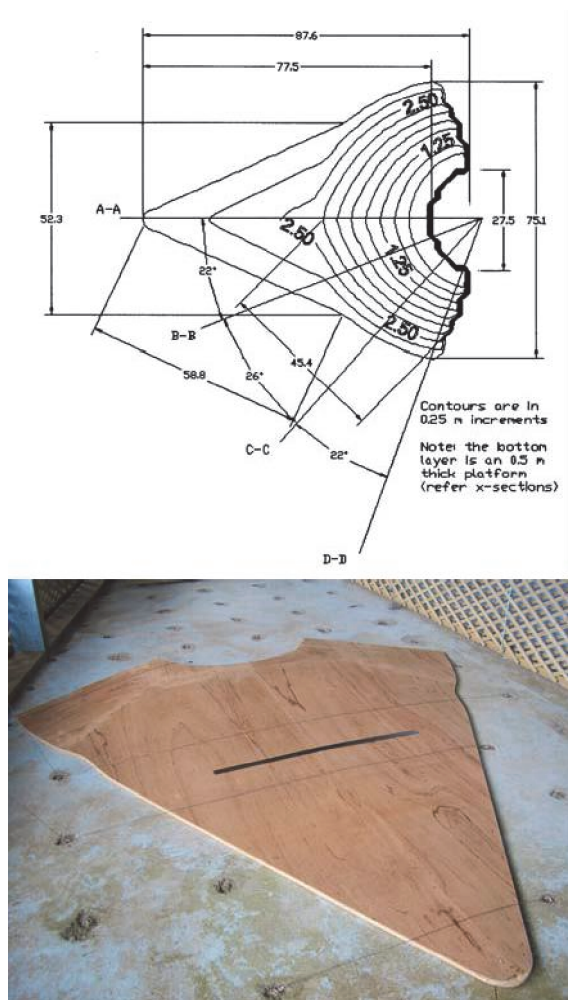


Figure 3. Plans of the final design were drawn to 1:20 scale (top) and a plywood reef was constructed for physical model testing (bottom).

The results of waves breaking on the 1:20 scale computer-designed reef (Figure 2) were very encouraging, with left and right hand waves peeling down either side of the reef with fast and hollow characteristics. However, due to the short width of the crest, the wave barreled and broke for only a short distance (~30 m when scaled up to full-size), the reef had ended and the wave had “fallen off” the end of the reef (Black et al., 2005). Modifications were undertaken to increase the width of the curving ‘wedge’ component at the back of the reef, which resulted in intense waves breaking the full-length of reef crest (Figure 4).

While the physical modelling provided very good evidence that the Mount Reef design would provide the quality of waves that it had been designed for, in reality, this shape could not be replicated with the materials and methods of the proposed construction – sand-filled mega-containers. Thus, the next stage of physical modelling considered using plastic tubing to replicate the shape of sand-filled mega-containers. Plumbing tubing could be bought of the shelf to replicate a variety of mega-container sizes at a scale of 1:21. Tubing was taped together to mimic the 2:1 ratio

of the mega-containers once they are filled with sand. An early version of the 1:21 mega-container reef model is shown in Figure 5.



Figure 4. Intense breaking right (top) and left (bottom) peeling waves on the modified computer-designed Mount Reef.

A series of tests and modifications were undertaken in order to ensure that a mega-container layout was developed that provided the best possible wave quality (e.g. Figure 6).



Figure 5. Version 2 of the 1:21 scale mega-container model of the Mount Reef design.

The final 1:21 scale model was next used in a number of simulations to assess the wave quality through a variety of wave and tidal conditions. 3 tide levels were tested (i.e. high, mid and low tide), and for each tide, 3 wave heights were tested, which were in the range 1-1½, 1½ -2 and 2-2½ meters. Above about 3 metres, the reef was forecast to close-out on the

surrounding sand banks, particularly at low tide, and so no tests needed to be done at that height (Black and Mead, 2005).



Figure 6. A 'clean' intense breaking wave on the final 1:21 scale mega-container model of the Mount Reef design.

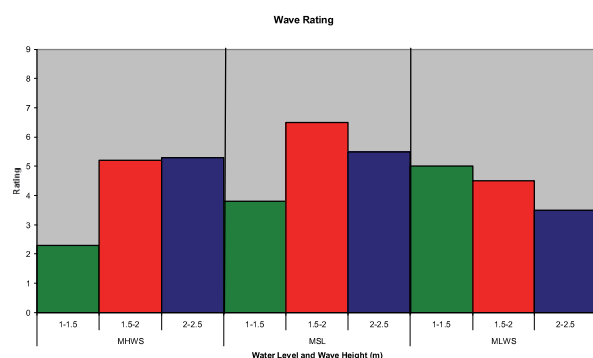


Figure 7. Physical modeling results of the assessment of the performance of the final reef design under differing wave heights and water levels. Each wave has been given a rating between 1 and 10 – (1 = Very Poor, 10 = Perfect Wave – non existent) (Moores et al, 2005)



Figure 8. Final mega-container layout for construction of the Mount Reef.

The wave quality during the tests was subjectively ranked from 1-10, which basically translated to 3

being crumbling or collapsing waves; 5 good waves; and 7 excellent and hollow waves. The results showed that the best waves were forecast to occur at mid tide on 1.5-2 m swells. These waves are ranked 6-7 which means that good surfers will get barreled (i.e. be able to ride inside the breaking crest of the wave, in the tube). The wave conditions are good at high tide for the clean swells around 1.5-2.5 m. Best waves close to low tide occur when the wave heights are less than 1.5 m.

The final physical model design was next converted into a container layout for construction (Figure 8).

4 Construction

The Mount Reef was designed primary to provide high-quality surfing waves during clean swell conditions (with prevailing south west winds, wave conditions on the north eastern New Zealand coast are often small and clean (due to the offshore winds). Previous multi-purpose reefs incorporating surfing amenity have utilised split-hull dredge and GPS to position the vessel and drop mega-containers to the seabed (e.g. Jackson, 2001). However, while this is a very cost effective method of multi-purpose reef construction, it is very difficult method for accurate placement the mega-containers. Gaps and humps between mega-containers disrupt the wave quality, especially at smaller wave heights, such as common at the Tay Street site in Mount Maunganui. The led to the development of the R.A.D. (Rapid Accurate Deployment) method of reef construction for the Mount Reef, which allows for the amalgamation of the constraints of construction with design details and tolerances (Black et al., 2005).

The RAD method is essential for minimisation of costs and risks. First, the reef is put together onshore and consists of the following components:

- An underlying web of customised sewn webbing
- A geomat (optional)
- Geo-containers attached on top
- Webbing straps for use during deployment

The Mount Reef was built in 2 separate halves, with 12 mega-containers on each half, ranging in size from 50 m long x 7.2 m wide x 3.2 m high (when filled) to 30 x 1.5 m wide x 1.0 m high (when filled). The twelve mega-containers were secured to the customised webbing, with a geomat underneath (to prevent differential settling), and then craned onto a barge (Figure 9).

Prior to deployment, manta-anchors were put into the seabed at precisely surveyed (i.e. an RTK system was used). The leader lines of the webbing holding the mega-containers together could then be threaded through the large eyelets in the heads of the manta-anchors so that the reef section could be winched into place.



Figure 9. Cranes with splitter-bars ready to load the first half of the Mount Reef onto a waiting barge.



Figure 10. The route taken by the barge to deploy each half of the Mount Reef (Source: Google Earth, 2006).

Once a clear weather window of at least 4 days was identified, the barge with the first half of the Mount reef on board was towed around to the reef site (Figure 10). The leader lines of the webbing were passed through the manta-anchors and the entire half of the reef (approximately 80 x 30 m) was winched to the seabed. The novel RAD method of reef deployment was completed in 3.5 hours without any unforeseen problems and filling of the containers commenced.

The mega-containers were filled in situ, with commercial divers inserting a custom-made fitting into filling ports and then an 8/6 Warman submersible pump extracting sand from the nearby seabed and pumping some 70-80 m into the containers. A long-neck digger sitting on a barge was used to both power and move the pump.

5 Reef Performance

At the time of writing, the Mount Reef is some 70% complete, with the main focus area still unfilled, a large mega-container to be placed and filled (presently leaving a 'gap' some 60 m long by 6 m wide through the middle of the right-hander (the replacement is

scheduled in the near future) and many of the containers not completely filled. Even so, the Mount Reef has been delivering the fast tubing waves it was designed to produce (Figure 11).



Figure 11. Top: Left-hand barrel on the Mount Reef, March 2006 (www.surf2surf.com). Bottom: One shot of a 17 shot sequence of a right-hand barrel on the Mount Reef, January 2007 (www.caughtonradar.com)

Evidence of the quality of the waves breaking on the Mount Reef have been regularly posted on the internet as both photographs and video clips (e.g. www.mountreef.co.nz and www.caughtonradar.com) these data and others are very useful for evaluating wave breaking performance and the design process. For instance, in October 2006, a significant gap between mega-containers was yet to be filled near the end of the right-hand section of the Mount Reef (Figure 12). The impacts on the wave face due to this gap were replicated during physical model testing (Figure 12). Several other components of both physical and numerical modelling have been verified with the actual results of waves breaking on the Mount reef during various phases of construction.

Continued research has included bi-monthly surveys of the reef, which have been used to calibrate sediment transport modelling (due to be published in a Scientific Journal later this year – Black et al., in prep), while the results of the waves breaking on the reef and the sand bars forming on the beach and around it have resulted in a large amount of numerical and physical modelling being validated and strongly confirms the design processes and research findings that have previously been published (e.g. Black and Mead, 2005, Moores et al, 2005; Mead and Black, 2001a, b).

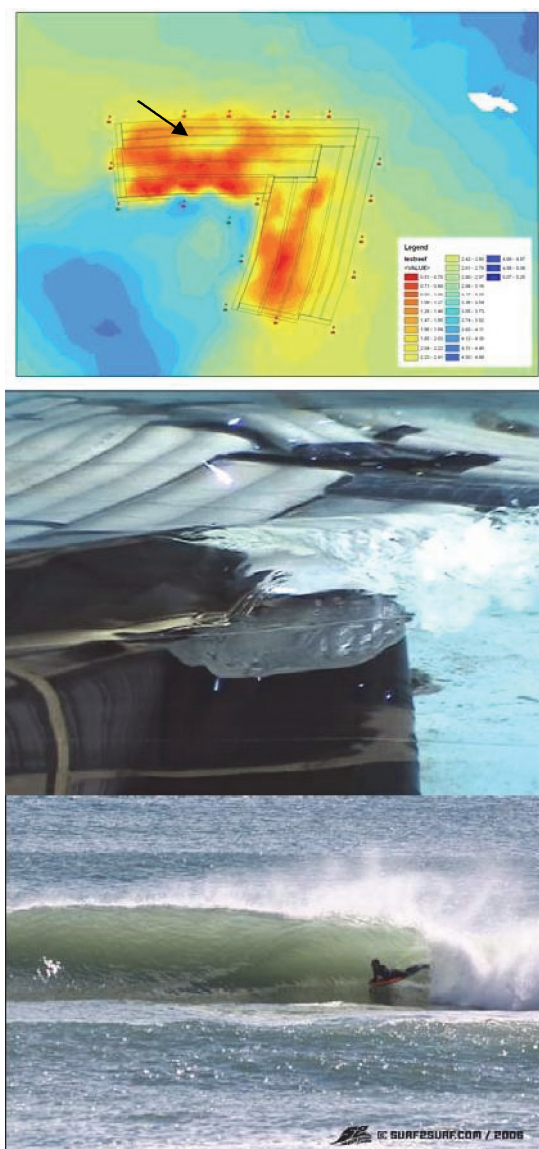


Figure 12. The top figure shows the results of the October 2006 bathymetry survey with the unfinished state of the reef obvious and a large gap indicated by an arrow. The middle figure shows the physical modelling results of a gap between containers and how it creates a ridge in the wave face. The bottom figure shows the impact of the gap in the reef on the wave face, i.e. a ridge up the wave.

6 Summary

It has been over 10 years since the Mount Reef was first considered, with the construction process now nearing completion. The detailed numerical modelling design was transformed through the used of scaled physical modelling in order to develop a layout for the sand-filled mega-containers that the reef is built of. The physical modelling amalgamated the constraints of construction methods and material with the detailed computer design. While not yet complete, waves breaking on the Mount Reef during favourable swell conditions have validated a large amount of numerical and physical modelling and strongly confirms the

design processes and research findings that have previously been published.

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